

1. Determinați aproximarea Padé de grad $[2,2]$ pentru $\sin(x)$ și $\cos(x)$

aproximarea Padé se definește ca $R_{2,2} = \frac{a_0 + a_1x + a_2x^2}{1 + b_1x + b_2x^2}$

P. $\sin(x)$

~~$\sin x$~~ Pentru această funcție avem nevoie de
primii $2+2+1=5$ termeni ai dezvoltării adăugăm
 $\sin(x) = x - \frac{x^3}{3!}$

$$= x^0 \cdot 0 + x^1 \cdot 1 + x^2 \cdot 0 - \frac{1}{3!} \cdot x^3 + 0 \cdot x^4$$

$$C_0 = 0 \quad C_1 = 1 \quad C_2 = 0 \quad C_3 = -\frac{1}{6} \quad C_4 = 0$$

afirm coeficienții b_1, b_2 rezolvând următorul sistem

$$\begin{bmatrix} C_2 & C_1 \\ C_3 & C_2 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} -C_3 \\ -C_4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -\frac{1}{6} & 0 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ 0 \end{bmatrix}$$

$$0 \cdot b_1 + b_2 \cdot 1 = \frac{1}{6} \Rightarrow b_2 = \frac{1}{6}$$

$$-\frac{1}{6}b_1 + 0 \cdot b_2 = 0 \Rightarrow b_1 = 0$$

afirm coeficienții lui a calculând:

$$a_j = \sum_{l=0}^j c_{j-l} b_l$$

$$a_0 = c_0 b_0 = 0 \cdot 1 = 0$$

$$a_1 = \sum_{l=0}^1 c_{1-l} b_l = c_1 b_0 + c_0 b_1 = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$a_2 = \sum_{l=0}^2 c_{2-l} b_l = c_2 b_0 + c_1 b_1 + c_0 b_2$$

$$= \cancel{A_2} 0$$

$$R_{2,2}(x) = \frac{x}{1 + \frac{x^2}{6}} \quad \sim \text{pentru } \sin x$$

cos x

~ Alegem primii 5 termeni din aproximarea clasică

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

$$= 1 \cdot x^0 + 0 \cdot x^1 - \frac{1}{2!} \cdot x^2 + 0 \cdot x^3 + \frac{1}{4!} \cdot x^4$$

$$c_0 = 1 \quad c_1 = 0 \quad c_2 = -\frac{1}{2} \quad c_3 = 0 \quad c_4 = +\frac{1}{4!} = +\frac{1}{24}$$

Calculăm matrice sistemul

$$\begin{bmatrix} c_2 & c_1 \\ c_3 & c_2 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} -c_3 \\ -c_4 \end{bmatrix}$$

$$\begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{24} \end{bmatrix}$$

$$\begin{cases} -\frac{1}{2} b_1 = 0 \\ -\frac{1}{2} b_2 = -\frac{1}{24} \end{cases} \quad (\Rightarrow) \quad \begin{cases} b_1 = 0 \\ b_2 = \cancel{\frac{1}{12}} \frac{1}{12} \end{cases}$$

Calculăm coeficienții a_0, a_1, a_2

$$a_0 = c_0 b_0 = 1 \cdot 1 = 1$$

$$a_1 = \sum_{l=0}^1 c_{1-l} b_l = c_1 b_0 + c_0 b_1 = 0 \cdot 1 + 1 \cdot 0 = 0$$

$$\begin{aligned} a_2 &= \sum_{l=0}^2 c_{2-l} b_l = c_2 b_0 + c_1 b_1 + c_0 b_2 = -\frac{1}{2} \cdot 1 + 0 \cdot 0 + 1 \cdot \frac{1}{12} = \\ &= -\frac{1}{2} + \frac{1}{12} \\ &= -\frac{5}{12} \end{aligned}$$

$$R_{2,2}(x) = \frac{1 - \frac{5x^2}{12}}{1 + \frac{x^2}{12}} \sim \text{pentru } \cos x$$